Steps:

<u>Exploratory:</u>

1. Take the IR spectra data, perform dim-reduction on them (such as PCA), and cluster. See if there's evidence that these aggregate into groups/subgroups.
2. Do the same thing with the percept vectors (the target for each odorant), and see if these form clusters. "Paint" the perceptual cluster labels onto the IR spectra clusters and see if there's a correspondence.

Prediction task:

1. Try a few models for classification, with odorant x feature matrix as your X, and the corresponding percept vectors as your y. Use cross-entropy as your loss function.
    a. The task is:
        i. Compute features for the molecules using Pyrfume (mordred features)
        ii. Throw away some features (feature selection: "select k-best")
        iii. Do the classification, and summarize the performance of the models (report scoring metrics and values. Try: precision, recall, F1. Also report AUROC/AUC. Show a confusion matrix. Set this up as a prediction task where you predict the single most dominant descriptor. Also try (if you like, if its easier) predicting pleasantness for each molecule.
2. Repeat this analysis on the spectral data alone (using spectral data as your X).
3. Repeat this analysis with spectral data concatenated onto the original features (physicochemical features)
4. Report: all AUROC/AUC/precision, recall scores across the different scenarios.